

Energy Harvesting-aware Contention Tree-based Access for Wireless Machine-to-Machine Networks

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Abstract—The Energy Harvesting-aware Contention Tree-based Access (EH-CTA) protocol is presented in this paper as a novel Medium Access Control (MAC) protocol for wireless Machine-to-Machine (M2M) networks where end-devices are equipped with energy harvesters. The protocol is theoretically modeled to analyze its performance. A performance comparison with an EH-aware Dynamic Frame Slotted-ALOHA (EH-DFSA) shows its superior performance. While EH-DFSA requires an estimation of the number of contenders per frame in order to dynamically adjust the frame length, EH-CTA uses short and fixed frame lengths. This ensures scalability and facilitates synchronization in highly dense M2M networks.

Index Terms—Machine-to-Machine networks; energy harvesting; medium access control; tree splitting algorithm.

I. INTRODUCTION

Two complementary strategies can be used to extend the lifetime of the end-devices in Machine-to-Machine (M2M) networks: *i*) reduce the energy consumption required for communications and *ii*) use energy harvesters that collect energy from the environment [1]. Towards this end, the Medium Access Control (MAC) protocol must take the variability of the energy-harvesting process into account, and use the energy available in an extremely efficient manner. This is the main motivation for the contribution presented in this paper, where a novel MAC protocol based on a Contention Tree Algorithm (CTA) is proposed for M2M networks with energy harvesting capabilities. As it has been shown in the literature [2], using tree-splitting algorithms can improve the performance of either static multi-user or random access protocols.

The performance of random access protocols and tree-splitting algorithms for M2M networks has been already studied in the past [3]–[5]. However, the analyses of the MAC layer with energy harvesting are rarely found. Previous works focus on slotted-ALOHA [6], CSMA [7] [8] and Dynamic Frame Slotted-ALOHA (DFSA) [9], [10]. However, to the best of our knowledge, the design and analysis of random access protocols with CTA for wireless networks with energy-harvesting has never received attention. This paper aims to fill this gap with the following contributions:

1) Design of a new MAC protocol, named Energy Harvesting-aware Contention Tree-based Access protocol (EH-CTA), suitable for M2M data collection networks where each end-device is equipped with an energy harvester and an energy-storage device (ESD).

2) Theoretical modeling of EH-CTA based on Markovian analysis to derive Key Performance Indicators and compute the performance of the network.

3) Comprehensive performance evaluation of EH-CTA and comparison with an Energy Harvesting-aware DFSA (EH-DFSA).

The remainder of this paper is organized as follows. The system model is described in Section II. The operation of EH-CTA is described in Section III. The analytical model is presented in Section IV. Section V discusses the accuracy of the analysis and the performance of the protocol. A comparison with EH-DFSA is shown in Section V. Section VI concludes the paper.

II. SYSTEM MODEL

A. Network and Data Model

We consider a single-hop wireless network composed of 1 coordinator (or gateway) and n end-devices in star topology. Each end-device is equipped with an energy harvester and an ESD which stores the energy captured by the harvester. The coordinator gathers data from the end-devices via periodic Data Collection Rounds (DCR), once every T_R seconds as shown in Figure 1. In each DCR, an end-device has exactly one new data packet ready to transmit to the coordinator.

A DCR is initiated when the coordinator broadcasts a Request for Data (RFD) packet. The RFD of the k -th DCR is followed by a sequence of F_k frames further divided into m slots as shown in Figure 1. The length of the data packets is fixed and the transmission time of a data packet is equal to the slot duration.

If an end-device has enough energy in its ESD at the beginning of a DCR, it enters into *active mode* and attempts to transmit its data packet according to the rules of the adopted MAC protocol. Otherwise, it remains in *sleep mode* and waits for the next DCR. The end-devices which are active transmit in randomly selected slots without performing Clear Channel Assessment. The coordinator transmits a feedback packet (FBP) at the end of each frame in order to inform about

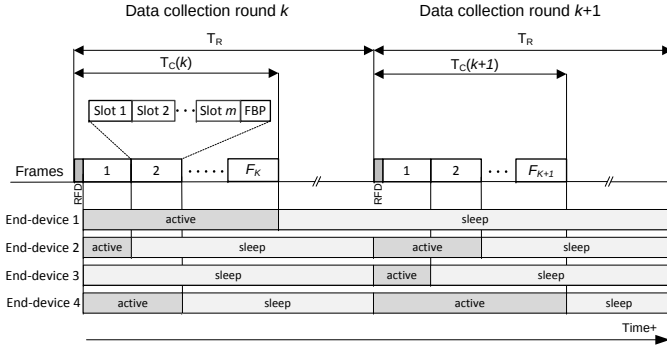


Figure 1: Example of Data Collection Process.

the status of each slot: (i) success, i.e., a data packet from an end-device is properly received and decoded; (ii) failure, i.e., one or more end-devices have transmitted in the slot but none data packet can be decoded by the coordinator due to a collision or channel error; or (iii) empty, i.e., the channel is idle. When an end-device succeeds in transmitting its data packet in a given frame, it enters into sleep mode and saves energy until the next DCR starts. On the contrary, if an end-device fails in a given frame, it will attempt to transmit its data packet in subsequent frames as long as it has enough energy available. If an end-device fails to transmit its data packet due to energy shortage in any given DCR, the data packet is discarded and not transmitted in subsequent DCRs.

The contention process finishes when all the end-devices have entered into sleep mode due to either successful transmission or energy shortage. We assume that the maximum time $T_c(k)$ elapsed since the k -th DCR starts until the contention process finishes is much shorter than the time between two consecutive DCRs, i.e., $T_c(k) \ll T_R$ for all k .

To focus on the MAC layer, it is assumed that all data packets are transmitted without errors induced by the wireless channel. In addition, we assume that none of the data packets involved in a collision can be decoded by the coordinator, i.e., there is no capture effect.

B. Energy Storage and Energy Consumption Models

The amount of energy stored in an ESD of an end-device can be modeled as a random variable which depends on the harvested energy and the energy consumed by the end-device throughout the DCRs. The energy stored in the ESD of the i -th end-device is denoted by $E_{ESD,i} \in \{0, 1\delta, 2\delta, \dots, N\delta\}$, where δ [Joule] is referred to as *energy unit*, and N is the normalized capacity of the ESD. The end-device enters into active mode if the energy in its ESD at the beginning of the k -th DCR, denoted by $E_{ESD,i}(k)$, is above a predefined energy threshold $E_{thr} = \varepsilon_{thr}\delta$, with $\varepsilon_{thr} \in \{0, 1, 2, \dots, N-1\}$. The probability that an end-device can take part in the k -th DCR is called activation probability, denoted by $p_{active}(k)$, which can be expressed as

$$p_{active}(k) = \Pr \{E_{ESD,i}(k) > E_{thr}\}. \quad (1)$$

Each time that an end-device transmits its data packet in a certain frame of a DCR it consumes a constant amount of energy, denoted by E_{tx} [Joules], which accounts for all the energy consumption in the communication phases with the coordinator: (i) the end-device transmits in 1 slot, (ii) it remains in sleep mode in the other $m-1$ slots, and (iii) it listens to the channel to receive the FBP. The energy consumed by an end-device in sleep mode is negligible. For mathematical tractability, we normalize the energy consumption E_{tx} to one energy unit, i.e., $E_{tx} = 1\delta$. Therefore, when the ESD is fully charged at the beginning of a DCR, i.e., it contains N energy units, the maximum number of data packet transmissions of an end-device is N .

C. Energy Harvesting Model

The energy harvester of the i -th end-device captures an amount of energy, denoted by $E_{H,i}(k)$, for the time interval T_R between two consecutive DCRs $(k-1)$ -th and k -th. The harvested energy $E_{H,i}(k)$ can be modeled as a discrete random variable with a probability mass function $q_j = \Pr \{E_{H,i}(k) = j\delta\}$ with $j \in \{0, 1, 2, \dots\}$ energy units, which depends on the characteristics of the energy source. The stochastic process $E_{H,i}(k)$ is considered to be identically and independently distributed (i.i.d) over all end-devices and DCRs.

The *energy harvesting rate*, denoted by $\bar{E}_H$, is defined as the average energy harvested by an end-device during the time T_R , which can be expressed as

$$\bar{E}_H = \mathbb{E} [E_{H,i}(k)]. \quad (2)$$

The dynamics of the energy harvesting process are considered to be slower than the contention process in a DCR. Therefore, the amount of energy that is harvested within the duration $T_c(k)$ of the k -th DCR is negligible and is not immediately available to be used during the DCR. Consequently, all the harvested energy $E_{H,i}(k)$ is ready to be used at the beginning of the next DCR.

III. ENERGY HARVESTING-AWARE CONTENTION TREE-BASED ACCESS PROTOCOL

In EH-CTA, the end-devices which become active in a DCR are organized into sub-groups using a tree-splitting algorithm. The algorithm can be represented with a *contention tree* where each node of the tree represents a time frame with a fixed number m of slots. The tree-splitting algorithm works as follows. In the first frame, all the active end-devices randomly select one slot to transmit their data packet. If two or more end-devices collide in a specific slot, a new frame is assigned only to the sub-group of end-devices that caused the collision in order to retransmit, and they are queued into a logical queue: the Collision Resolution Queue (CRQ). Therefore, if there are k slots with collision in one frame of level d of the tree, with $d \in \{1, 2, \dots, N\}$, then k new frames are scheduled in level $d+1$, and k sub-groups of end-devices are queued into the CRQ. The process leads to the formation of a tree whose expansion stops at frames with only successful and/or empty

slots, meaning that all collisions have been resolved. In any case, the process is stopped at a maximum level N , which is the normalized capacity of the ESD and thus limits the number of levels where an end-device can retransmit. An end-device transmits its data-packet in one frame of every level d of the tree until it succeeds in one level or enters in energy shortage. The CRQ is represented at each end-device by two integer numbers: (i) the position of the end-device in the CRQ, and (ii) the total length of the CRQ, i.e., the number of subgroups of end-devices that are waiting for retransmission. The coordinator broadcasts the status of the m slots and the length of the CRQ in the FBP at the end of each frame. Using this information, an end-device that has collided in a given frame can compute its position in the CRQ. Once an end-device has entered in the CRQ, it will retransmit its data packet in the next frame if it occupies the first position in the CRQ; otherwise, it just waits its turn without contending during successive frames. The position of an end-device in the CRQ is always decremented by one at the end of each frame. Consequently, an end-device only needs to receive the FBP of those frames where it contends in order to know whether it has succeeded or has collided.

IV. ANALYSIS

A. Performance Metrics

The *probability of delivery* is defined as the probability that an end-device becomes active in the k -th DCR and succeeds in transmitting its data packet to the coordinator.

The *time efficiency* is defined as the ratio between the number of data packets successfully transmitted to the coordinator in the k -th DCR, and the total number of slots required to complete the DCR.

B. Markov Chain Model

The aim of this analysis is to compute the performance metrics in steady state conditions. Towards this end, we need to evaluate the steady-state probability distribution of the energy available in the ESDs at the beginning of a DCR. By properly setting to a high enough value the energy threshold for an end-device to become active, it is possible to assume that the number of end-devices that fall in energy shortage during a DCR is negligible. Then, the probability that an end-device succeeds in one frame of any level of the contention tree basically depends on the number of active end-devices at the beginning of the DCR, the number of slots and the level number. Consequently, we can evaluate the steady-state probability distribution by analyzing the evolution of the energy of a single ESD, which is an approximation that neglects the interactions among the ESDs of different end-devices.

The evolution of the energy available in an ESD can be modeled with the discrete-time Markov chain shown in Figure 2. Each state in the chain is defined by $\{e(t), d(t)\}$, where $e(t) \in \{0, 1, \dots, N\}$ is a stochastic process which represents the number of energy units stored in the ESD at time t ; and $d(t) \in \{0, 1, \dots, N\}$ is a stochastic process which represents that either an end-device is in sleep mode when

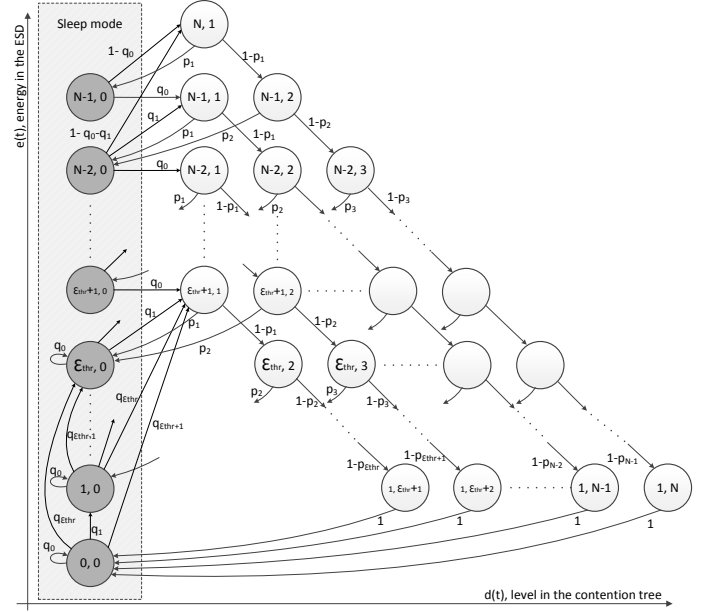


Figure 2: Markov chain that models the evolution of the energy available in an ESD. Some transitions are omitted for ease of understanding.

$d(t) = 0$, or the level number in the contention tree in which an end-device contends when $d(t) \in \{1, \dots, N\}$. State transitions in the chain do not occur at fixed time intervals. The Markov chain is characterized by a transition matrix $P = [p_{ij}]$, where p_{ij} is the one-step transition probability defined as

$$\Pr \{e(t+1) = e_j, d(t+1) = d_j | e(t) = e_i, d(t) = d_i\} \quad (3)$$

An end-device that has either succeeded or entered in energy shortage in a DCR remains in sleep mode (i.e., in one of the states with $d_i = 0$) until the next DCR. At the beginning of a DCR, the energy harvested ε_H in the last T_R interval is added to the energy available in the ESD, i.e., $e_j = e_i + \varepsilon_H$. The probability that an end-device harvests $\varepsilon_H \in \{0, 1, 2, \dots\}$ energy units is denoted by q_{ε_H} . Then, if the energy available in the ESD is below or equal a threshold ε_{th} , i.e., $e_j \in \{0, 1, \dots, \varepsilon_{th}\}$, the end-device makes a transition from state $(e_i, 0)$ to state $(e_j, 0)$ and continues in sleep mode. On the contrary, an end-device becomes active if $e_j \in \{\varepsilon_{th} + 1, \dots, N\}$, making a transition from state $(e_i, 0)$ to state $(e_j, 1)$. Note that e_j is limited to the capacity N of the ESD.

The transition probability from state $(e_i, 0)$ to any state (e_j, d_j) at the beginning of a DCR is

$$p_{ij} = \begin{cases} q_{\varepsilon_H}, & \text{if } (e_j \leq \varepsilon_{th}) \text{ and } (d_j = 0) \\ q_{\varepsilon_H}, & \text{if } (\varepsilon_{th} < e_j < N) \text{ and } (d_j = 1) \\ 1 - \sum_{k=0}^{N-1-e_i} q_k, & \text{if } (e_j = N) \text{ and } (d_j = 1) \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

When an end-device becomes active, it retransmits in one frame of every successive level of the contention tree until either it succeeds or its ESD falls in energy shortage. If the

end-device succeeds in one frame of level $d_i \in \{1, \dots, N\}$, it makes a transition from state (e_i, d_i) to state $(e_i - 1, 0)$ for $e_i \in \{1, 2, \dots, N\}$. If the end-device does not succeed, there are two possible transitions: (i) to state $(e_i - 1, d_i + 1)$ when the end-device has enough energy to retransmit in one frame of the next level, i.e., $e_j \in \{1, 2, \dots, N - 1\}$; or (ii) to state $(0, 0)$ when the end-device falls in energy shortage, i.e., $e_j = 0$.

The transition probability from state (e_i, d_i) to state (e_j, d_j) with $d_i \in \{1, 2, \dots, N\}$ is

$$p_{ij} = \begin{cases} p_d, & \text{if } (e_j = e_i - 1) \text{ and } (d_j = 0) \\ 1 - p_d, & \text{if } (e_j = e_i - 1) \text{ and } (d_j = d_i + 1) \\ 1, & \text{if } (e_i = 1) \text{ and } (e_j = 0) \text{ and } (d_j = 0) \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

where p_d is the probability that an end-device succeeds in transmitting its data packet in one frame of level $d = d_i$. This probability is derived in the next subsection.

C. Probability of Success in one Frame and Energy Threshold

The probability that an end-device succeeds in transmitting its data packet in one frame of level d can be expressed as

$$p_d = \left(1 - \frac{1}{m}\right)^{n_d - 1} \quad \text{for } d \geq 1, \quad (6)$$

where m is the number slots per frame and n_d is the number of end-devices which contend in one frame of level d .

In the first frame of a steady-state DCR, i.e., in level $d = 1$, the number n_1 of end-devices that contend is equal to the average number of end-devices that become active, which can be expressed as $n_1 = n \cdot p_{active}^{SS}$, where n is the total number of end-devices and p_{active}^{SS} is the activation probability in steady-state, i.e., for large index k of DCR.

The value of the energy threshold ε_{th} must be set to guarantee that an end-device that becomes active in a DCR has enough energy to contend until success. We assume that the value of ε_{th} is set to ensure that the number of active end-devices that fall in energy shortage during a DCR is negligible. Under this assumption, the value of n_d for $d > 1$ can be derived as follows. First, the probability that k of n_d end-devices transmit in the same slot of a frame can be calculated as

$$p_s(k) = \binom{n_d}{k} \left(\frac{1}{m}\right)^k \left(1 - \frac{1}{m}\right)^{n_d - k}, \quad (7)$$

and the average number of empty, success, and collision slots in that frame can be calculated as $S_d^E = m \cdot p_s(0)$, $S_d^S = m \cdot p_s(1)$, and $S_d^C = m - S_d^E - S_d^S$, respectively.

If there are S_d^C slots with collision in one frame of level d , then $F_{d+1} = S_d^C$ new frames are scheduled in level $d+1$. Each new frame in level $d+1$ is assigned only to the sub-group of end-devices that caused a collision in the same specific slot of level d . The average number of end-devices that succeed in one frame of level d , denoted by n_d^S , is equal to the average number S_d^S of slots with success. Therefore, the average number of end-devices that collide in one frame of level d , denoted by n_d^C , can be calculated as $n_d^C = n_d - S_d^S$. Since we assume

that the number of end-devices that fall in energy shortage in a DCR is negligible, the n_d^C end-devices will contend again in F_{d+1} new frames of level $d+1$. Consequently, the average number n_{d+1} of end-devices that contend in one frame of level $d+1$ can be approximated by

$$n_{d+1} \simeq \frac{n_d^C}{F_{d+1}} = \frac{n_d - n_d \left(1 - \frac{1}{m}\right)^{n_d - 1}}{m - m \left(1 - \frac{1}{m}\right)^{n_d} - n_d \left(1 - \frac{1}{m}\right)^{n_d - 1}}. \quad (8)$$

In order to set the appropriate value for the energy threshold ε_{th} in EH-CTA, we need to calculate the average number of frames where an end-device has to contend until it succeeds, denoted by $\mathbb{E}[d]$, which can be expressed as

$$\mathbb{E}[d] = \sum_{d=1}^{\infty} d \cdot p_d \cdot \prod_{i=1}^{d-1} (1 - p_i). \quad (9)$$

The energy threshold needs to be adjusted to the value of $\mathbb{E}[d]$, i.e., $\varepsilon_{th} \approx \mathbb{E}[d]$, depending on the number of end-devices and slots. For example, for a network of 1000 end-devices, the value of ε_{th} must be close to 5, 4, or 3 energy units when m is 5, 10, or 20 slots, respectively.

D. Steady-State Probability Distributions

As it can be observed in Figure 2, when $p_d > 0$ for $d \in \{1, \dots, N\}$, $q_0 > 0$ and $q_1 > 0$, the Markov chain is aperiodic and irreducible, and thus ergodic. Therefore, it admits a unique steady-state probability distribution, denoted by $\pi = [\pi_{e,d}]$, which can be expressed as

$$\pi_{e,d} = \lim_{t \rightarrow \infty} \Pr \{e(t) = e, d(t) = d\}, \quad (10)$$

and satisfies that $(P' - I)\pi' = 0$, where P is the transition matrix and I is the identity matrix. This condition can be solved for π by calculating the eigenvector of P' that corresponds to an eigenvalue equal to 1. Recall that the transition matrix P depends on p_d (6), which also depends on p_{active}^{SS} . On the other hand, p_{active}^{SS} can be expressed from the steady-state probability distribution of the energy available in the ESD at the beginning of a DCR, denoted by $\pi^B = [\pi_{e,d}^B]$, as follows

$$p_{active}^{SS} = \pi_{\varepsilon_{thr}+1,1}^B + \dots + \pi_{N,1}^B = \sum_{e=\varepsilon_{thr}+1}^N \pi_{e,1}^B. \quad (11)$$

Since an end-device is in sleep mode before a DCR starts, π^B can be expressed as $\pi^B = \pi^S P$, where $\pi^S = [\pi_{e,d}^S]$ is the steady-state probability distribution conditioned on being in sleep mode, which is calculated as

$$\pi_{e,d}^S = \begin{cases} \frac{\pi_{e,0}}{\sum_{i=0}^{N-1} \pi_{i,0}}, & \text{if } (d = 0) \\ 0, & \text{if } (1 \leq d \leq N) \end{cases}. \quad (12)$$

Finally, we compute the steady-state probability distributions as follows. Firstly, we build the transition matrix P by setting the steady-state activation probability to a test value of 0, i.e., $p_{active}^{SS-test} = 0$. Secondly, we compute π , π^S , and π^B . Thirdly, we compute the analytical value of p_{active}^{SS} (11)

by using π^B . And finally, we check the relative error between the test and analytical values of the activation probability. These steps are repeated iteratively by increasing $p_{active-test}^{SS}$ until the error is below 0.1%, which indicates that the results obtained for π , π^S , and π^B are correct.

E. Probability of Delivery

Once the steady-state probability distribution π^B of the energy available in the ESD at the beginning of a DCR is computed, we can formulate the expression to calculate the steady-state *probability of delivery* for EH-CTA as follows

$$p_{delivery}^{SS} = \sum_{d=1}^{\varepsilon_{thr}} \sum_{e=\varepsilon_{thr}+1}^N \pi_{e,1}^B \prod_{k=1}^{d-1} (1-p_k) p_d + \sum_{d=\varepsilon_{thr}+1}^N \sum_{e=d}^N \pi_{e,1}^B \prod_{k=1}^{d-1} (1-p_k) p_d. \quad (13)$$

Recall that an end-device which becomes active in a steady-state DCR starts in one of the states $(e, 1)$ with probability $\pi_{e,1}^B$ for $e \in \{\varepsilon_{thr} + 1, \dots, N\}$. Then, the end-device retransmits its data packet in one frame of every level d until the packet is successfully decoded by the coordinator with probability p_d . The end-device fails if it enters in energy shortage before the packet is successfully decoded.

F. Time Efficiency

The *time efficiency*, denoted by η_t , can be formulated as the average number of slots with successful data packets in a DCR divided by the total number of slots in a DCR, which can be expressed as

$$\eta_t = \frac{\sum_{d=1}^N \mathbb{E}[m_{S,d}]}{\sum_{d=1}^N \mathbb{E}[m_d]} = \frac{S_1^S + \sum_{d=2}^N S_d^S \prod_{i=2}^d S_{i-1}^C}{m + \sum_{d=2}^N m \prod_{i=2}^d S_{i-1}^C}, \quad (14)$$

where $\mathbb{E}[m_d]$ is the average number of contention slots in level $d \in \{1, \dots, N\}$, $\mathbb{E}[m_{S,d}]$ is the average number successful slots in level d , m is the number of slots per frame, S_d^S is the average number of successful slots in a frame of level d , and S_d^C is the average number of collided slots in a frame of level d . Recall that in EH-CTA if there are S_d^C slots with collision in one frame of level d , then S_d^C new frames are scheduled in level $d+1$, and thus the average number of frames in level d can be calculated as $\prod_{i=2}^d S_{i-1}^C$.

In the next section, we validate the analytical model proposed to calculate the steady-state probability of delivery and time efficiency for EH-CTA.

V. PERFORMANCE EVALUATION

A. Scenario

We consider a wireless network formed by 1 coordinator surrounded by a number n of end-devices in the transmission range of the coordinator. Each end-device includes an energy harvester and an ESD with $N = 10$ energy units of capacity. We assume that the energy harvested by an end-device during

the time between two consecutive DCRs follows a binomial distribution with probability mass function

$$q_j = \binom{N_H}{j} \left(\frac{\bar{E}_H}{N_H} \right)^j \left(1 - \frac{\bar{E}_H}{N_H} \right)^{N_H-j} \quad (15)$$

for $j \in \{0, 1, 2, \dots, N_H\}$, where $N_H = 10$ is the maximum number of energy units that can be captured by an energy harvester, and $\bar{E}_H \in [0 \dots N_H]$ is the EH rate.

Results for EH-CTA have been obtained analytically and through computer-based simulations using MATLAB where the actual procedure of the protocol is executed for each node. The tight match between analytical and simulation results validate the accuracy of the analytical model proposed in Section IV. The results of 1000 simulation samples have been averaged for each test case.

The performance of EH-CTA has been compared with that of the DFSA protocol [10] for data collection networks with energy harvesting (also referred to as EH-DFSA). In EH-DFSA, the number of slots per frame is dynamically adjusted to the estimated number of end-devices that contend in each frame. An end-device that becomes active in a DCR randomly selects one of the slots in every frame to transmit its data packet. When an end-device either succeeds or falls in energy shortage, it enters into sleep mode and stops contending again in subsequent frames. Otherwise, it reattempts transmission of its data packet as long as it has enough energy to do so. At the end of each frame, the coordinator estimates the number of end-devices that will contend in the next frame and broadcasts the number of slots in the FBP. Results for EH-DFSA have been obtained through computer-based simulations.

B. Results

The probability of delivery and the time efficiency are represented in Figure 3 and Figure 4, respectively, as a function of the number m of slots per frame. They have been evaluated by considering: $n \in \{100, 1000\}$, $\bar{E}_H \in \{0.25, 0.5, 2, 4, 8\}$, and $N = 10$. In EH-CTA, the energy threshold has been set to $\varepsilon_{thr} = 6$ for $m < 5$, $\varepsilon_{thr} = 5$ for $m = 5$, $\varepsilon_{thr} = 4$ for $5 < m \leq 10$, and $\varepsilon_{thr} = 3$ for $10 < m \leq 40$. Since the frame length of EH-DFSA is dynamically adjusted, the results for EH-DFSA are constant over the number of slots per frame, which only applies to EH-CTA.

Results show that the probability of delivery for EH-CTA increases when the number of slots per frame increases. This is due to the fact that the probability p_d that an end-device succeeds in one frame of level d of the contention tree increases when m increases, thus leading the average number of retransmissions and the energy consumed per end-device to decrease, and the probability of delivery to increase. The probability of delivery for EH-CTA and EH-DFSA increases with the EH rate. Indeed, the higher the number of energy units available in the ESDs, the higher the number of end-devices that become active in a DCR and the higher the number of possible packet retransmissions. In addition, if the total number of end-devices increases, the probability of delivery for EH-CTA is degraded because the probability of collision is

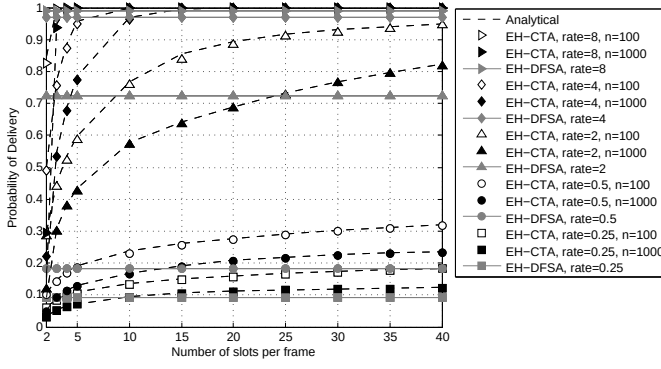


Figure 3: Probability of delivery over the frame length in EH-CTA for different harvesting rates and number of end-devices.

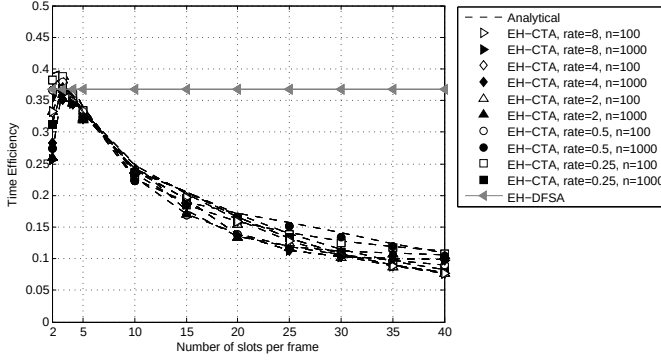


Figure 4: Time efficiency over the frame length in EH-CTA for different harvesting rates and number of end-devices.

higher. However, the probability of delivery for EH-DFSA is insensitive to the number of end-devices because the number of slots is equal to the number of contenders in every frame.

Figure 3 shows that EH-CTA can outperform EH-DFSA in terms of probability of delivery, for any EH rate and number of end-devices, if the number of slots per frame in EH-CTA is properly adjusted. For example, if $n = 100$ and $\bar{E}_H \in \{0.25, 0.5, 2, 4, 8\}$, then the value of m in EH-CTA must be equal or greater than 4, 5, 8, 7 and 3 slots, respectively; and if $n = 1000$ and $\bar{E}_H \in \{0.25, 0.5, 2, 4, 8\}$, then the value of m in EH-CTA must be equal or greater than 10, 15, 25, 10 and 4 slots, respectively.

Figure 4 shows that the time efficiency for EH-CTA is maximized for $m = 3$ slots, $\eta_t \approx 0.38$, and it is degraded as the number of slots per frame increases. Indeed, while the number of frames required to resolve the contention in EH-CTA tends to its minimum value as the number of slots increases, the delay (in slots) is minimized for $m = 3$ slots regardless of the number of end-devices that contend in the DCR [5]. In addition, EH-CTA slightly outperforms EH-DFSA in terms of time efficiency at $m = 3$ and it is very similar for different EH rates and number of end-devices. There is a trade-off between probability of delivery and time efficiency for EH-CTA. When the number of slots per frame increases, more end-devices can eventually succeed in transmitting data

to the coordinator in a DCR, thus increasing the probability of delivery at the cost of reducing the time efficiency and the data collection rate. However, as it can be observed in Figure 3, with low (e.g., 0.25) and high (e.g., 8) EH rates, EH-CTA can be configured with a very low number of slots, at almost no cost in the probability of delivery, and increase the time efficiency to a certain value close to the maximum. However, with intermediate EH rates (e.g., between 0.5 and 4 energy units), EH-CTA must be configured with a number of slots per frame which depends on the total number of end-devices in the network and the harvesting rate.

VI. CONCLUSIONS

The energy harvesting-aware Contention Tree-based Access (EH-CTA) protocol has been presented and theoretically modeled in this paper. There is a trade-off between the probability of delivery and the time efficiency. The time efficiency for EH-CTA is maximized for 3 slots per frame. However, if the number of slots is high, the time efficiency decays and the probability of delivery increases. Results show that EH-CTA can outperform EH-DFSA in terms of probability of delivery, for any given energy harvesting rate and number of end-devices. Furthermore, while EH-DFSA requires to estimate the number of contending end-devices in every frame to adapt the frame length, EH-CTA uses a constant frame length. While the frame length of EH-CTA can be very short, the frame lengths of EH-DFSA must be as high as the number of contenders in every frame, thus causing scalability problems as the number of end-devices increases. Future work aims at including transmission errors and capture effect in the analysis presented in this paper.

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